KPI Dashboards & Business Insights

Data Today. Better Decisions Tomorrow.

Turn your data into meaningful insights.
Track performance. Make better decisions.

Track Performance

Find Insights

Support Decisions

Drive Business Growth

> “Good dashboards don't just show numbers — **they tell a story.**”

KPI Dashboards & Business Insights

Turn data into meaningful insights. Help managers understand performance and make better decisions.

From Data to Decisions

Analyze. Visualize. Decide.

The Manager's Request

"Just give me a dashboard. I want to know how the business is performing."

- 10,000 sales transactions
- 50 products
- 12 months of sales
- 8 sales regions
- Customer information

The Question

- Should you show 10 charts?
- 20 numbers?
- Every product?
- Every transaction?
- A beautiful pie chart?

Not necessarily.

What Should Be on the Dashboard?

A good dashboard should help answer:

"What is happening in the business, and what should I do about it?"

Key Performance Indicators (KPIs)
Show the most important numbers (e.g., total sales, revenue growth, profit, number of customers).

Trends Over Time
Show how the business is performing across months.

Breakdowns
Show performance by product category, region, or customer segment (only the key ones).

Actionable Insights
Highlight what's working, what's not, and where to focus.

Guidelines for a Good Dashboard

1. **Keep it focused**
   Show only the most important information.
2. **Use clear visuals**
   Charts and KPIs that are easy to understand.
3. **Highlight trends**
   Show how performance changes over time.
4. **Provide comparisons**
   Include previous period or target (e.g., last month, last year).
5. **Show key breakdowns**
   Product, region, or customer segment (top performers).
6. **Make it actionable**
   The dashboard should help the manager take decisions.

Key Takeaway:

A dashboard is not just a collection of charts — it's a tool for better decisions.

Right Insights
Right Actions
Business Growth

First: What Is a KPI?

KPI = Key Performance Indicator

A KPI is a number that helps you understand whether something important is, tlophoning well dashboats.

A number with context tells a story.

Measure. Compare. Improve.

Example: College Café

If you run a college café, what numbers would you care about?

- Daily sales
- Number of customers
- Average order value
- Best-selling product
- Customer complaints

These could become KPIs.

Example: Online Clothing Store

Your sales last month were:

₹10 lakh

"Yes! ₹10 lakh sounds great."

But wait. Last month the sales were:

₹15 lakh

"Now ₹10 lakh doesn't look so great."

Numbers Need Context

A number by itself doesn't tell the whole story. You need context.

- **Current month vs previous month**
  Is performance improving or declining?
- **Current year vs previous year**
  Are we growing over the long term?
- **Actual sales vs target**
  Are we meeting our goals?
- **Other relevant comparisons**
  By product, region, customer segment, etc.

Key Takeaway:

KPIs aren't just numbers. They help us interpret performance and make better decisions.

Better Data
Better Insights
Better Decisions

KPI Identification

Choose the right KPIs for your sales dashboard.

Not everything is a KPI. Choose what matters.

Right KPIs. Clear Insights. Better Decisions.

The Manager's Request

"Create a sales dashboard."

What does the manager actually need to know?

Focus on the most important information, not everything in the dataset.

Useful KPIs for a Sales Dashboard

Total Sales
How much did we sell?

Total Profit
How much did we actually earn after costs?

Number of Orders
How many transactions happened?

Average Order Value
How much does a customer spend per order?

Sales Growth
Are sales increasing or decreasing?

Top Product
Which product is selling the most?

Regional Sales
Which region is performing best?

Key Points to Remember

1. **Understand the purpose**
   What decisions does the manager need to make?
2. **Focus on key KPIs**
   Select the most relevant metrics.
3. **Keep it simple**
   Avoid unnecessary numbers and charts.
4. **Make it actionable**
   The dashboard should help answer "what should we do next?"
5. **You can add more later**
   Start with key KPIs and expand if needed.

Key Takeaway:

The right KPIs help you tell the story of the business and support better decisions.

Measure
Understand
Act
Grow

Not Every Number Is a KPI

A dashboard is not a data dump. It should show what matters.

Data tells you what exists. KPIs tell you what matters.

Focus. Simplify. Drive Decisions.

Example Dataset (Columns)

#	Column Name	Type
1	Customer ID	Text
2	Customer name	Text
3	Phone number	Text
4	City	Text
5	Product	Text
6	Quantity	Number
7	Price	Number
8	Sales	Number
9	Cost	Number
10	Profit	Number
11	Date	Date

Can you put all of these on the dashboard?

Technically, yes. Should you? **No.**

- A dashboard is not a data dump.
- Too much information creates confusion.
- Focus on the numbers that help in decision-making.
- Not all columns are KPIs.
- Some columns are important for analysis but don't need to be on the dashboard.

Example KPIs for a Sales Dashboard

Total Sales
How much did we sell?

Total Profit
How much did we earn after costs?

Number of Orders
How many transactions happened?

Average Order Value
How much does a customer spend per order?

Sales Growth
Are sales increasing or decreasing?

Top Product
Which product is selling the most?

Regional Sales
Which region is performing best?

Key Takeaway:

Choose KPIs, not everything in the dataset. Show what matters, not just what exists.

Better Insights
Better Decisions
A More Effective Dashboard

Let's Think Like an Executive

You have 10 seconds. What do you want to know?

Right Insights.
Faster Decisions.
Stronger Business.

From Data
to Decisions.

Key Questions for an Executive

Are sales going up or down?
How is the business performing over time?

Are we hitting our target?
How do actual results compare to goals?

Which products are driving revenue?
What are our best and worst performing products?

Which region is underperforming?
Where should we focus more attention?

Is profit increasing along with sales?
Are we growing profitably?

Important Rule

Start with business questions, not charts.

Don't start with:
"I want to create a pie chart."

Start with:
"What decision should this dashboard help someone make?"

Key Takeaway:

A good dashboard answers business questions and helps make better decisions.

See the Big Picture
Find Opportunities
Drive Action

Choosing the **Right Chart**

Choose a chart based on what you want to communicate, not just because it looks attractive.

Right data. Right chart. Clearer insights.

Simple Charts. Stronger Decisions.

1 Show a Trend

Example:
Monthly sales from January to December

Use a Line Chart

Because you're interested in how something changes over time.

2 Compare Categories

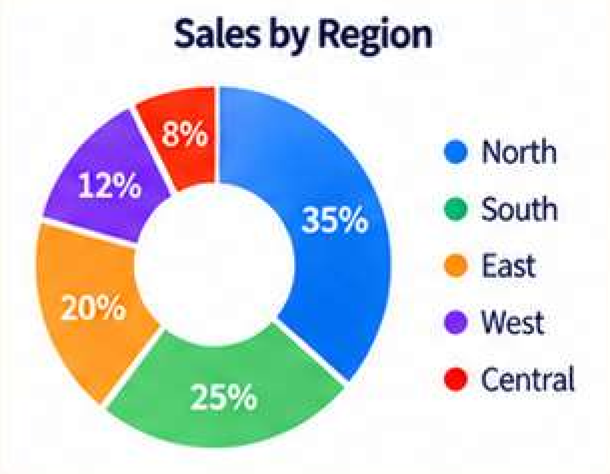
Example:
Sales of Product A, B, C, D

Use a Bar/Column Chart

Because you're comparing different categories.

3 Show a Part of a Whole

Example:
How total sales are divided across regions



Use a Pie/Donut Chart

Useful for showing proportions. But if you have many categories (e.g., 15 regions), a bar chart may be clearer.

4 Show One Key Number

Example:
Total Sales = ₹25 lakh

Total Sales

₹25 lakh

▲ +12%
vs last month

Use a KPI Card

You don't need a chart for every number.

Additional Tips

- ✓ Choose the right chart for the message.
- ✗ Don't use a chart just because it looks attractive.
- ✓ Keep it simple and easy to read.
- ✓ Avoid pie charts with too many categories.
- ✓ Use clear titles, labels, and colors.
- ✓ Make sure the chart helps answer a business question.

Key Takeaway:

The best chart is the one that clearly communicates the right message to the right audience.

Better Visuals
Better Insights
Better Decisions

Quick Question

Imagine you're showing:

Monthly revenue from January to December

Which would you choose?

Think about what you want to communicate.

Right Chart. Clearer Story. Better Decisions.

Which chart makes the most sense?

A Pie chart

Revenue by Month

B Line chart

Monthly Revenue

C Donut chart

Revenue by Month

D KPI card

Total Revenue

₹25 lakh

▲ +12%

vs last month

Correct Answer: B. Line chart

A line chart is best for showing a **trend over time**, such as monthly revenue from January to December.

Key Takeaway:

Choose a chart based on what you want to communicate, not just what looks attractive.

Better Charts
Better Insights
Better Decisions

Another Question

You want to compare:

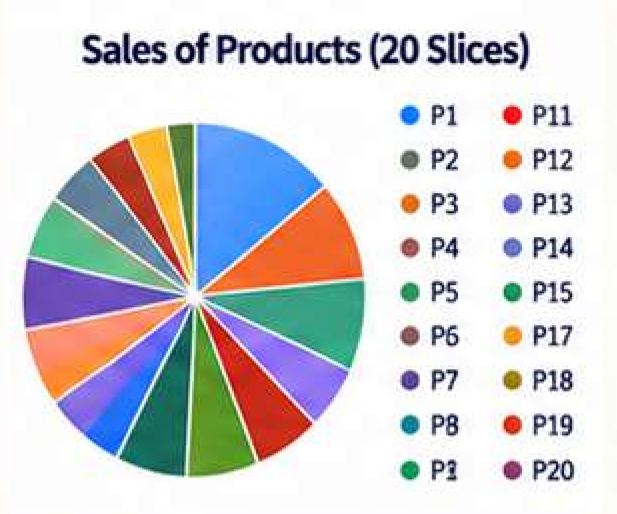
Sales of five products.

Which is more appropriate?

A Bar chart

B Line chart

C Pie chart with 20 slices



D Random chart because it looks cool

The right chart makes comparison easy to understand.

Choose Smart. Visualize Better. Make Better Decisions.

Correct Answer: A. Bar chart

A bar chart is best for **comparing different categories**, such as sales of five products.

Key Takeaway:

Choose a chart based on what you want to communicate, not just what looks attractive.

Clearer Visuals
Easier Comparisons
Better Decisions

Dashboard Design

A dashboard is not about more charts. It's about clearer insights.

Less clutter.
More clarity.
Better decisions.

Good Design.
Better Insights.
Real Impact.

Bad Dashboard (Too Cluttered)

- 12 colors
- 15 charts
- 8 fonts
- 20 numbers
- 5 pie charts
- Hard to understand

Good Dashboard (Clear & Focused)

Dashboard Design Principles

- Make important information easy to find.
- Use a clean and consistent color scheme.
- Keep it simple and focused.
- Use a visual hierarchy.
- Show insights, not just data.
- Answer business questions.

Key Takeaway:

A good dashboard is simple, clear, and focused on what matters.

Better Dashboards
Better Insights
Better Decisions

Don't Make Everything Equally Important

Some numbers need more attention than others.

Use visual hierarchy to highlight what matters.

Same Data.
Better Focus.
Smarter Decisions.

Which number needs immediate attention?

Sales	Profit	Orders	Growth
₹50 lakh	₹8 lakh	3,200	-18%
▲ +5% vs last month	▲ +2% vs last month	▲ +6% vs last month	vs last month

Growth: -18% immediately deserves attention.

Sales may look impressive, but the negative growth tells us something has changed.

Why It Matters?

- Numbers need context.
- Not all metrics are equally important.
- Use visual hierarchy (size, color, placement) to highlight key insights.
- Help the viewer focus on what needs action.

Key Takeaway:

A good dashboard guides attention to what matters most.

Clearer Insights
Faster Decisions
Greater Impact

Another Example – Average Can Mislead

Don't just look at the number. Understand the story behind the data.

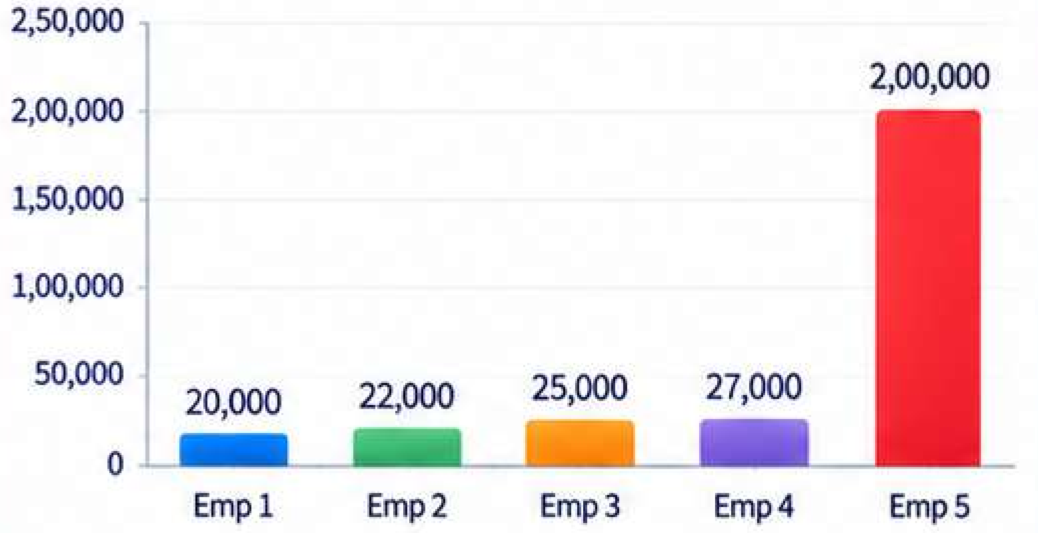
Better Analysis. Better Insights. Better Decisions.

Averages can be affected by extreme values. Always look at the data behind the numbers.

Employee Salaries

Employee	Salary (₹)
Employee 1	20,000
Employee 2	22,000
Employee 3	25,000
Employee 4	27,000
Employee 5	2,00,000

Employee Salaries (Bar Chart)



Key Insight

The average salary of ₹58,800 looks high, but it doesn't reflect the salary of most employees.

Average Salary

$$\frac{(20{,}000 + 22{,}000 + 25{,}000 + 27{,}000 + 2{,}00{,}000)}{5}$$

= ₹58,800

What's Happening?

- Four employees earn between ₹20,000 – ₹27,000.
- One employee earns ₹2,00,000.
- This one high value increases the average to ₹58,800.
- So the average **does not represent what most employees earn.**

What Else Can You Look At?

- Median salary (middle value)
- Salary distribution (bar chart or box plot)
- Minimum and maximum salaries
- Understand outliers in the data

Key Takeaway:

Don't just look at the number. Understand the data behind it.

Deeper Analysis
Better Insights
Smarter Decisions

AI Can Help Build Dashboards

Use AI to think through the dashboard structure before you start building.

AI won't replace your thinking. It will make you a better analyst.

Better Questions. Better Insights. Better Dashboards.

1 Your Data (Excel)

Date	Product	Region	Sales	Quantity
2024-01-05	Laptop	North	75,000	10
2024-01-12	Mobile	South	50,000	8
2024-01-20	Tablet	East	30,000	5
2024-01-28	Laptop	West	60,000	7
2024-02-03	Mobile	North	55,000	6
...	...	...	...	...

2 Your Prompt to AI

Analyze this sales dataset and suggest an executive KPI dashboard.

Identify:

1. The 5–7 most useful KPIs
2. The business question answered by each KPI
3. The best chart type for each important comparison or trend
4. Useful filters or segments
5. Potential anomalies or data-quality issues

Do not invent information that is not present in the dataset.

3 AI Response (Example Output)

1. **Recommended KPIs**
   - **Total Sales** – How much did we sell?
   - **Total Profit** – How much did we earn?
   - **Number of Orders** – How many transactions?
   - **Average Order Value** – How much per order?
   - **Sales Growth** – Are sales increasing?
   - **Top Product** – Which product sells the most?
   - **Regional Sales** – Which region performs best?
2. **Chart Suggestions**
   - **Sales Trend:** Line chart (monthly sales)
   - **Product Comparison:** Bar chart
   - **Regional Sales:** Bar or map chart
   - **KPI Cards:** For key numbers (Total Sales, Profit, etc.)
3. **Useful Filters**
   - Date range, Region, Product, Customer segment
4. **Potential Issues**

   Missing values in some rows, outliers in sales, inconsistent date formats

4 Executive Dashboard (Example)

Key Takeaway:

AI helps you think through the dashboard, so you can build smarter, not just faster.

AI + Your Business Knowledge = Better Decisions

Same Data. Better Questions. Greater Insights.

AI-Generated Business Insights

Good insights go beyond what the chart shows. They help answer "why" and "so what".

From data to insights. From insights to better decisions.

Ask Better Questions. Get Deeper Insights. Create Real Impact.

Monthly Sales Trend

Basic AI Insight (Just repeats the chart)

"Sales increased significantly in March, indicating strong business performance."

Why it's not useful?

- It only repeats what the chart shows.
- It doesn't explain the reason.
- It doesn't help in decision-making.

Better AI Insight (Goes deeper)

"March sales increased 18%, primarily because Product A sales increased 35% and the North region recorded its highest monthly sales."

Why it's more valuable?

- Explains what contributed to the change.
- Provides context (product and region).
- Helps in decision-making (e.g., replicate success, investigate further).

Key Takeaway:

Good insights explain the reason behind the numbers, not just the numbers themselves.

Deeper Insights
Smarter Decisions
Greater Impact

The Three-Level Insight

Go beyond the numbers. Move from observation to action.

Same data.
Deeper thinking.
Bigger impact.

From Data
to Decisions.
With AI.

1 Level 1 — What happened?

Understand the key change in the data.

"Sales increased 18%."

This tells us **what** changed.

2 Level 2 — Why might it have happened?

Find the possible reasons behind the change.

"Product A sales increased significantly."

This helps us understand **why** it might have happened.

3 Level 3 — What should we do?

Turn insight into action.

"Consider increasing inventory and marketing support for Product A."

Inventory
Increase stock to meet higher demand.

Marketing
Allocate more budget to promote Product A.

Monitor
Track performance in the coming months.

This is where business thinking begins. It helps us decide **what to do next**.

From insights to action!

Key Takeaway:
Great analysis goes beyond **what happened**. It explains **why**, and suggests **what to do next**.

Better Questions
Deeper Insights
Smarter Decisions

AI Shouldn't Invent the "Why"

AI can suggest possible reasons, but it should not state them as facts.

Ask questions.
Check the data.
Don't assume.
Think critically.

Better Analysis.
Better Questions.
Better Decisions.

Example

Your dashboard shows:

Sales increased 20%.

AI says:

"Sales increased because customers responded positively to the new advertising campaign."

Can we conclude that?
Not necessarily.

AI may be guessing. It should not invent a reason without evidence from the data.

There could be many reasons for the increase:

Prices changed
Higher or lower prices can impact sales.

A competitor had supply problems
Customers may have switched to your product.

A seasonal event occurred
Festivals, holidays, or special seasons can increase demand.

A new product launched
A successful product launch can drive sales.

One large customer placed an order
A single big order can significantly increase sales.

Important Rule

" Correlation is not automatically causation."

If two things happened at the same time, that doesn't prove one caused the other.

Just because two things happen together doesn't mean one caused the other.

What to Do Instead?

- Look at the data in detail.
- Compare multiple factors.
- Ask further questions.
- Consider alternative explanations.
- Use domain knowledge and business context.

Key Takeaway:

Use AI for possibilities, not assumptions.
Validate insights before drawing conclusions.

Same Data.
Deeper Thinking.
Smarter Decisions.

Main Activity – Build Your KPI Dashboard (Part 1)

Create a one-page executive dashboard from the sales dataset.
But don't start by opening the chart menu. Start with questions.

Good dashboards start with good questions.

Analyze.
Visualize.
Gain Insights.
Make Better Decisions.

1 Your Dataset (Sample)

You receive a sales dataset with the following columns:

Date	Product	Region	Quantity	Sales	Cost	Profit
2024-01-05	Laptop	North	10	75,000	60,000	15,000
2024-01-12	Mobile	South	8	50,000	38,000	12,000
2024-01-20	Tablet	East	5	30,000	22,000	8,000
2024-01-28	Laptop	West	7	60,000	45,000	15,000
2024-02-03	Mobile	North	6	55,000	40,000	15,000
...	...	...	...	...	...	...

Your actual dataset may contain more rows.
Do not change or invent data.

2 Your Task

Create a one-page executive dashboard using this dataset.

Goal:
Present the most important information clearly and visually.

Your dashboard should:

- Show key KPIs (e.g., total sales, profit, orders).
- Include relevant charts (trend, comparison, composition).
- Use suitable filters or segments (e.g., date range, product, region).
- Be clear, simple, and easy to understand.
- Focus on what matters for business decision-making.

3 Think First — Key Questions

Before creating charts, think about:

1. **What are the most important KPIs for this business?**
   (e.g., total sales, profit, number of orders, growth)
2. **What business questions should this dashboard answer?**
   (e.g., Which products are performing best?)
3. **What chart type is best for each comparison or trend?**
   (e.g., line chart, bar chart, pie/donut)
4. **What filters or segments will be useful?**
   (e.g., date range, product, region)
5. **Are there any anomalies or data-quality issues to be aware of?**
   (e.g., missing values, unusually high/low sales)

Key Takeaway:

Start with questions, not charts.
A great dashboard is built with **purpose.**

Ask.
Analyze.
Visualize.
Create Business Impact.

Build Your KPI Dashboard (Part 1) – Step by Step

From questions to a one-page executive dashboard.

Good dashboards start with questions, not charts.

Ask. Plan. Visualize. Get Insights.

1 Ask Business Questions

Start by understanding what the manager needs to know.

1. How much did we sell?
2. How much profit did we make?
3. Is sales performance improving?
4. Which products perform best?
5. Which region performs best?
6. Are there any unusual changes?

2 Choose Your KPIs

Select 5–7 KPIs maximum. Choose the ones that answer your business questions.

- Total Sales
- Total Profit
- Total Orders
- Average Order Value
- Sales Growth
- Best Product
- Best Region

Tip: Don't choose KPIs just because they sound nice. Choose what's relevant.

3 Choose Your Visuals

Decide which information needs a chart.

Information	Best Chart Type
Monthly sales trend	Line chart
Sales by product	Bar chart
Sales by region	Bar chart (or map)
Total sales	KPI card
Total profit	KPI card
Sales by product category	Bar chart / Pie chart
Sales growth	KPI card or small trend chart

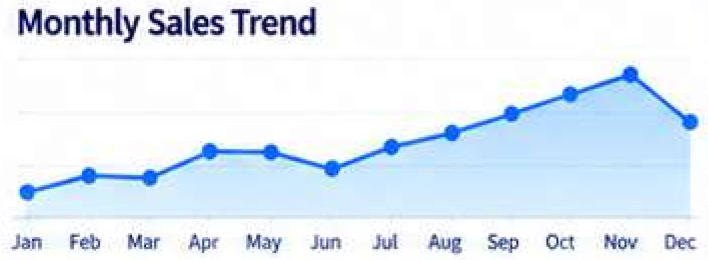
Tip: Use the simplest chart that clearly answers the question.

4 Build the Dashboard

Keep everything on one page. Use a clean structure.

Sales Dashboard Jan 2024 – Dec 2024

Total Sales	Total Profit	Total Orders	Avg Order Value
₹5,40,000	₹1,20,000	1,250	₹432
▲ +12%	▲ +8%	▲ +15%	▲ +6%



5 Ask AI for Insights

Once your dashboard is ready, ask AI for deeper analysis.

Example Prompt:

Analyze the following KPI results.

Identify:

1. Three important business observations
2. Two unusual or potentially concerning patterns
3. Two questions that require further investigation
4. Three possible business actions

Use only the information provided.
Do not invent causes or unsupported explanations.
Clearly distinguish observations from hypotheses.

Key Takeaway:

A great dashboard is built with purpose – start with questions, then choose the right KPIs, visuals, and insights.

Better Questions.
Better Dashboards.
Better Decisions.

Dashboard Quality Check

Before showing your dashboard to the manager, ask yourself these questions.

A good dashboard is clear, accurate and useful.

Check.
Validate.
Improve.
Deliver with confidence.

1 Can someone understand the dashboard quickly?

Key information should be easy to find at a glance.

2 Are the KPIs actually relevant?

Choose KPIs that answer real business questions.

3 Are the charts appropriate?

Use the right chart type for the data and comparison.

4 Are the numbers accurate?

Verify calculations, totals, and data sources.

5 Is there enough context?

Include time period, units, filters, and key notes.

6 Have I accidentally created a misleading visual?

Check scales, labels, and design choices.

7 Can I explain every number?

Be prepared to explain how each number is derived.

8 Can this dashboard help someone make a decision?

If not, rethink your dashboard.

Key Takeaway:

A good dashboard isn't just about showing data – it's about helping people make better decisions.

Clearer Insights.
Better Decisions.
Greater Impact.

The "Beautiful but Useless" Dashboard

A dashboard isn't a poster. It's a decision-support tool.

Looks good is not enough. It must help make decisions.

From data display to real impact.

Pretty Dashboard (Looks Great, But...)

Visually impressive, but doesn't clearly support a business decision.

- Looks modern and attractive
- Lots of charts and colors
- Shows many numbers
- No clear business question
- Too much information
- Hard to find the key message
- Doesn't directly support a decision

VS.

Useful Dashboard (Supports Decisions)

Simple, focused, and answers key business questions.

- Answers key business questions
- Clear and simple design
- Relevant KPIs only
- Easy to understand
- Helps identify trends and top performers
- Provides useful insights
- Supports decision-making
- Looks professional and clean

A useful dashboard turns data into action.

Key Takeaway:

Don't just make your dashboard look good. Make it useful.

Less Clutter.
More Clarity.
Better Decisions.

Final Challenge – Look Beyond the Headlines

Good dashboards don't just show what's going well — they also highlight what needs attention.

Higher sales doesn't always mean higher profit.

Same Data. Deeper Thinking. Better Decisions.

Key Metrics

Sales	Profit	Orders	Total Cost
₹50 Lakh	₹3 Lakh	1,250	₹47 Lakh
▲ +20%	▲ +5%	▲ +15%	▲ +35%
vs. previous period	vs. previous period	vs. previous period	vs. previous period

Key Observation

Sales and orders increased, but costs increased by 35%, so profit grew only slightly.

At first glance, the business looks successful. But higher costs are reducing profitability.

Sales vs. Profit Trend

Monthly Costs

Orders Trend

What Looks Good?

- Sales increased by 20%.
- Orders increased by 15%.
- Business activity is growing.

What Needs Attention?

- Costs increased by 35%.
- Profit grew only slightly (5%).
- Profit margin may be under pressure.

Key Question

Would you report "strong business growth" or investigate profitability first?

A good dashboard should encourage you to ask the right questions.

Key Takeaway:

Don't just look at individual numbers — look at the full story.

More Questions.
Better Analysis.
Stronger Decisions.

Same Data, Different Dashboards, Different Value

It's not about how many charts you have. It's about what decisions it helps you make.

Simplicity Drives Action.

Less Noise. More Clarity. Better Decisions.

Dashboard A

Beautiful, but overloaded. Looks impressive, but hard to find the key message.

- Too many charts and colors
- No clear focus
- Hard to quickly understand the key message
- Looks professional, but not actionable
- Executives may get confused
- Decoration does not add value

Dashboard B

Simple, focused, and actionable. Shows what matters.

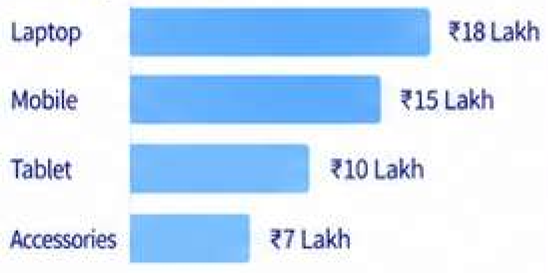
Sales Performance Dashboard

Jan 2024 – Dec 2024

Total Sales	Total Profit	Total Orders	Avg. Order Value	Profit Margin	Total Cost
₹50 Lakh	₹3 Lakh	1,250	₹4,000	6%	₹47 Lakh
▲ +20%	▲ +5%	▲ +15%	▲ +4%	▼ -2%	▲ +35%
vs. previous period	vs. previous period	vs. previous period	vs. previous period	vs. previous period	vs. previous period

Monthly Sales & Profit Trend

Sales by Product



Key Insights

1. Sales increased by 20% compared to the previous period.
2. Costs increased by 35%, which limited profit growth.
3. Laptop and Mobile together contributed 66% of total sales.

Recommended Action

- Investigate cost increase and identify key drivers.
- Focus on high-performing products (Laptop & Mobile).
- Explore ways to improve profit margin.

Which one for the CEO?

Dashboard B

Executives don't need more decoration. They need answers.

Executives want to know:

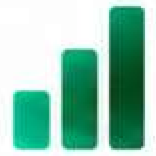
1. What happened?
2. Why does it matter?
3. What should we do?

> A dashboard isn't a poster.
> It's a **decision-support tool.**

Remember This Framework

Turn data into decisions.

Data – Collect and prepare → **KPI** – Choose what matters → **Visualization** – Use the right charts → **Insight** – Find key takeaways → **Decision** – Take action



Less Decoration.
More Decisions.
Greater Impact.